

Adaptive Design of Clustered Surveys and Experiments



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Introduction and Motivation

State Problem

Algorithms and Performance Guarantees

Inference

Application

Conclusion

- ▶ Design of clustered surveys/experiments is given by **two numbers**: number of clusters (K , **width**) and units per cluster (N , **depth**)
 - ▶ select (K, N) to **minimize variance** of a population mean or ATE
 - ▶ subject to a **hard budget**, **variable cost per unit**, and **fixed cost per cluster**—sunk before recruitment like travel or staff training

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- ▶ Optimal design is known since Cochran (1977) but depends on the **ICC**
 - ▶ in practice, designers **guess** ...
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 - ▶ ... **guessing is difficult**, ICC varies an order of magnitude within a single population (Knox and Chondros, 2004)

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 - ▶ ... **guessing is difficult**, ICC varies an order of magnitude within a single population ([Knox and Chondros, 2004](#))
- ▶ Adaptive designs are enticing but they **mess up inference**
 - ▶ why minimize estimator's variance if we cannot use standard inference?

- ▶ **Research questions**

- ▶ Can we design optimal clustered experiments under unknown ICC?
- ▶ Can we do so without compromising inference?

- ▶ Two **pilot-based algorithms** learn the ICC while the experiment runs

- ▶ induces negligible excess variance, beats any static design, and is rate unimprovable

- ▶ Under **error symmetry**, **adaptivity does not compromise inference**

- ▶ **general theory** of inference invariance which includes Neyman allocations

- ▶ **End to end illustration**

- ▶ simulations, revisit Fink et al. (2020, AER)—up to **41% lower variance** at equal cost—and implement a **live survey** using the algorithm

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- ▶ Extends mechanically to experimental design
- ▶ Nested **random effects** model [**A1**]

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \quad \text{Var}(u_k) = \tau^2, \quad \text{Var}(\varepsilon_{ki}) = \sigma^2, \quad u_k \perp\!\!\!\perp \varepsilon_{ki}$$

- ▶ **unknown variance ratio** $\lambda_0 = \sigma^2/\tau^2$, one-to-one with ICC = $1/(1 + \lambda_0)$
 - ▶ $\omega = (\tau^2, \lambda_0) \in \text{int}(\Omega)$ with Ω **known** and **compact** [**A2**]
- ▶ **Goal**

$$\min_{K,N} \text{Var}(\hat{\mu}_{K,N}) = W(K, N) \quad \text{st} \quad K(F + VN) \leq B \quad \text{a.s.}$$

- ▶ F fixed cost per cluster, V variable cost per unit

$$N^*(\lambda_0) = \sqrt{\lambda_0 F / V}, \quad K^*(\lambda_0) = \frac{B}{F + V N^*(\lambda_0)}$$

- Assume **interior solution** for $\lambda \in [\lambda_m, \lambda_M]$ [**A3**]

$$W_0 = W(K^*(\lambda_0), N^*(\lambda_0)) = \Theta(1/B), \quad \Delta_\pi = W(K^\pi, N^\pi) - W_0$$

- **Cost of guessing** (Lemma 5.2)

$$\Delta_{\text{guessing}} = W(K^*(\lambda), N^*(\lambda)) - W_0 = \Theta(1/B) \text{ for all } \lambda \neq \lambda_0$$

- **same order as the oracle variance itself**
- true for **any** static input, including the minimax λ input (van Breukelen and Candel, 2015)

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► **Pilot-based policies**

$$\text{pilot } (K_p, N_p) \implies \hat{\lambda}_p = \hat{\lambda}_{K_p, N_p} \implies (K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$$

- clusters and units cannot be *unsampled*
- **implementable** pilots: $K_p \leq K^\pi$ and $N_p \leq N^\pi$ a.s. $\implies$ **circularity**

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- ▶ **Why committing early?**

- ▶ suppose $K_p = 20$ and $K^*(\hat{\lambda}_p) = 50 \implies$ additional clusters are valuable for λ estimation

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► Why committing early?

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► **ACS**: fixes shallow $N_p = N^*(\lambda_m)$, samples **one cluster at a time**, stops before **oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$

► nearly **all exploration inside the pilot**, at no additional sampling cost

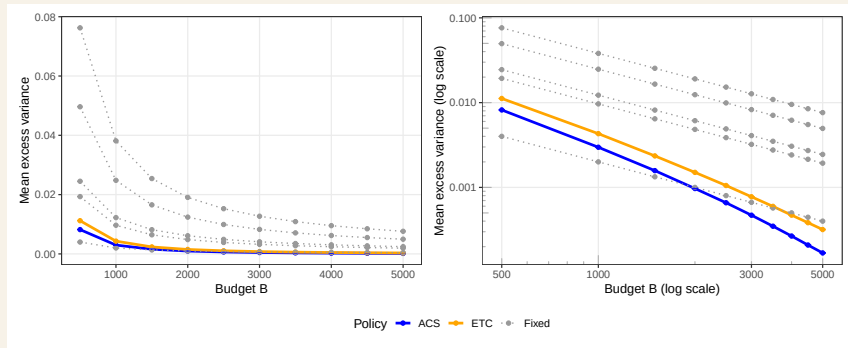
Theorems 5.4 and 5.5. Upper and Lower Bounds

Under $[A1 - A3] + \text{subgaussian } u_k, \varepsilon_{ki}$ $[A4]$, there exist constants $C_\pi < \infty$, $\perp B$,

$$\mathbb{E} [\Delta_{\text{EtC-CS}}] \leq \frac{C_{\text{EtC-CS}}}{B^2}, \quad \mathbb{E} [\Delta_{\text{ACS}}] \leq \frac{C_{\text{ACS}}}{B^2}$$

Moreover, for every policy π there exists $\omega \in \Omega$ such that $\mathbb{E}_\pi[\Delta_\pi] \geq C/B^2$.

- ▶ **asymptotically negligible** excess variance, $O(1/B^2) \ll \Theta(1/B)$
- ▶ algorithms asymptotically **dominate every misspecified static policy**
- ▶ **minimax rate optimal**, matching bounds



Notes: Linear scale (left) and Log Scale (right). Static policies imputed at $\lambda \in \{1, 10, 30, 60, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

- **Low excess variance** with ACS outperforming EtC-CS
- **Faster convergence**, slopes ≈ -2 (EtC-CS and ACS), ≈ -1 (static)

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- ▶ (K^π, N^π) **depends on pilot data** $\Rightarrow$ is $\hat{\mu} \mid (K^\pi, N^\pi)$ centered or asympt normal?
 - ▶ adaptive sampling biases estimates and destroys normality (Nie et al., 2018; Wager, 2024)
- ▶ **Adjustment tradition:** restore normality by **reweighting** the observations
 - ▶ Hadad et al. (2021); Zhang et al. (2021); Yin and Shen (2005)
- ▶ **Invariance tradition:** under **exact normality** designs driven by **sample variances** do not disturb inference on means
 - ▶ Stein (1945); Rinott (1978); Dudewicz and Dalal (1975)

- ▶ This paper builds on the **invariance tradition** and shows that under **error symmetry**, conditional inference is safe when the design depends on the data only through **quasi-even** functionals
- ▶ extends the invariance principle in three ways
 - ▶ **normality** → **symmetry** of the errors
 - ▶ **sample variance** → any **asymptotically even** functional of the data
 - ▶ **fixed-stage** adaptivity → **arbitrary** sequential designs
- ▶ Theory extends **well beyond clusters**, allows for conditioning of **non-degenerate events** (Hahn et al., 2011) and avoids **pilot data exclusion** (Viviano, 2022)

Theorem 6.10. Design-Robust Inference for EtC-CS and ACS

Under $[A1 - A3] + (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots) \stackrel{d}{=} -(u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ [A5], for any (non-degenerate) terminal design event A_B satisfying an anti-concentration condition,

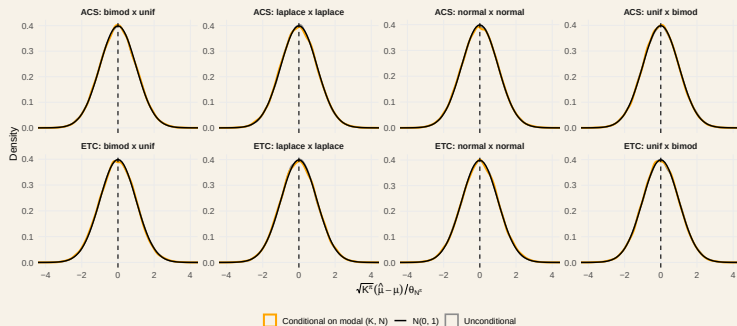
$$\mathcal{L} \left(\frac{\sqrt{K^\pi} (\hat{\mu}_{K^\pi, N^\pi} - \mu)}{\hat{b}_{K^\pi, N^\pi}} \mid A_B \right) \Rightarrow N(0, 1) \quad \pi \in \{\text{EtC-CS}, \text{ACS}\}$$

- **Nothing needs to change:** full sample, usual clustered SEs and normal CV—no reweighting, no sample splitting, no discarded data
- Theory extends to any **asymptotically even policy**, including **Neyman allocations** with pilot variances (and PS as in Hahn et al. (2011))

- ▶ Under symmetry, cluster errors **factor** into

signs $\times$ magnitudes with signs $\perp\!\!\!\perp$ magnitudes

- ▶ Even functionals of the data are functions of **magnitudes alone**
 - ▶ conditioning on an even design **freezes** the design and the magnitudes...
 - ▶ ...but leaves the signs **untouched**, so a **CLT runs through the signs**
- ▶ **Sharp boundary**
 - ▶ reading **odd** functionals (interim means, futility rules) or **asymmetric** errors restores the usual pathologies



Notes: Empirical density at the modal design. Distribution of u_k (left), of ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2,000$, $F = 20$, $V = 2$.

- Conditional and unconditional laws **coincide**;
- **Boundary is real**, odd policies or skewed errors introduce **not vanishing** distortions

Policy Reads...	Assumption on the Errors	
	Gaussian	Symmetry (New)
Levels	✗	✗
Residuals	✓	✗
Squared Errors (even)	✓	✓
Squared Residuals (asympt even)	✓	✓

Table: Policies and Error Assumptions

Scale	Does Surrogate Coupling Survive?	Non-degenerate Events?	Tools
Unit	✗	—	—
$1/B$	✓	✗	Anscombe
$1/\sqrt{B}$ (new)	✓	✓	Symmetry and Evenness

Table: Scale of Conditioning Events

- ▶ We can condition wherever surrogate coupling survives
- ▶ The symmetry $\times$ evenness theory allows for conditioning on **non-degenerate** events
- ▶ Unconditional inference does **not** require symmetry (Anscombe)

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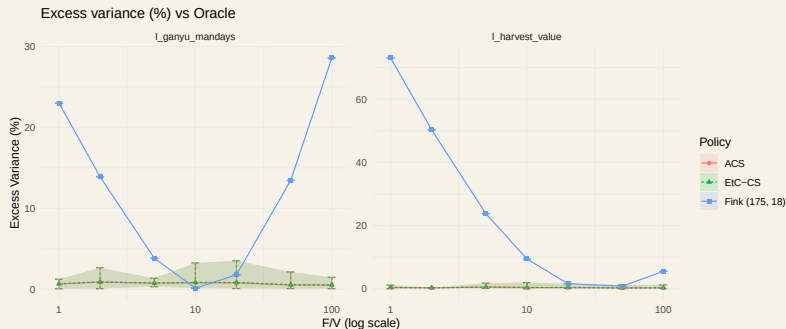
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- ▶ **Setting:** loans to farmers in Zambia with treatment at the **village** level
 - ▶ $K = 175$ villages $\times N \approx 18$ households
- ▶ **Was the realized design cost-optimal?**
 - ▶ do not observe cost structure, but possibly $F/V \in [1, 2]$
 - ▶ their design is only rationalizable at $F/V \approx 33$
 - ▶ away from this ratio, the realized design carries **up to 70% excess variance**



Notes: Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

- ▶ Algorithms remove nearly all excess variance uniformly across cost-ratios
- ▶ Not claiming that FJM got it wrong—simply **no design is safe without cost-aware learning**
- ▶ **Symmetry is testable**, and not rejected here, so standard inference applies

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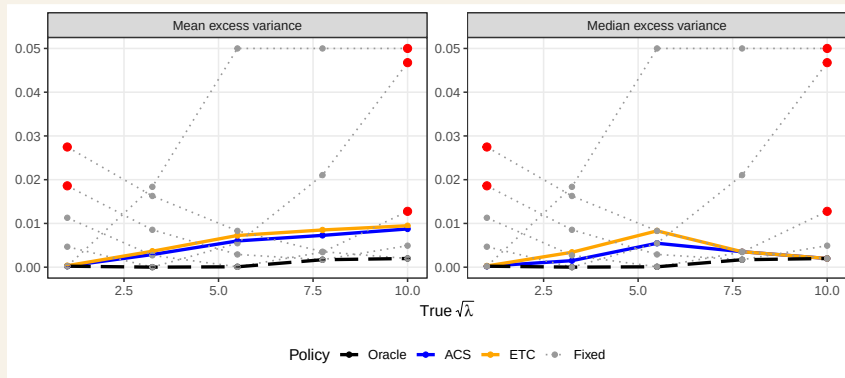
- ▶ **Framework** for cost-optimal clustered designs under **unknown ICC**, casting the classical width-vs-depth trade-off into a **sequential learning** problem
- ▶ Two **adaptive algorithms**
 - ▶ excess variance $O(1/B^2)$, negligible against the $\Theta(1/B)$ oracle variance
 - ▶ rate unimprovable, and dominate any misspecified static design $\Theta(1/B)$
- ▶ **Adaptivity does not compromise inference under symmetry**
 - ▶ even designs leave estimators asymptotically normal
 - ▶ no corrections, no discarded data
 - ▶ general principle which validates many popular sampling rules
- ▶ **Economically meaningful**
 - ▶ 41% lower variance in FJM at equal cost,
 - ▶ runs live in the field

Thank you!

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Carlos Gonzalez-Perez — and I am on the market!

- ▶ **In practice**, people fix a design and plug-in ICC guesses
- ▶ **Survey theory**: identifies the ICC as central but (almost) never estimates it **inside** the design problem (Cochran, 1977; Bethel, 1989)
- ▶ **Statistics and Biostatistics**
 - ▶ **Sample-size re-estimation** with nuisance parameters (Stein, 1945)
 - ▶ re-branded as **internal pilots** (Wittes et al., 1999)
 - ▶ ICC re-estimation (Lake et al., 2002; van Schie and Moerbeek, 2014) no cost structure, no optimality theory, a single margin (width or depth)
 - ▶ **Static robust** designs over ICC uncertainty, either Bayesian (Spiegelhalter, 2001) or **maximin** (van Breukelen and Candel, 2015)
 - ▶ adaptive learning dominates by an order of magnitude
- ▶ First paper to establish optimality guarantees against the oracle under unknown ICC in the joint allocation problem



Notes: Static policies imputed at $\lambda \in \{1, 10, 30, 60, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

- Adaptive **worst case beats every static worst case** (minimax included), static policies win only near their guess and only at finite budgets